SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR.



Predictive Analytics

“ Discovering Crop Patterns and Co-Cultivation Relationships in Agriculture Using Association Rule Mining”

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1. Problem Statement

1.1 Background and Motivation

Agriculture is the main source of livelihood for a large share of the population in India, and the choices made on farms every season, such as which crop to sow, in which season and on how much land, shape both farm incomes and national food supply. These choices are influenced by climate, soil, water availability, market prices, tradition and the experience of neighbouring farmers. Over many years, state and district authorities have recorded the outcome of these choices in the form of crop statistics: for every district, year, season and crop, the area cultivated and the quantity produced are noted.

Such records accumulate into very large tables. In practice, however, they are mostly used for totals, averages and year-to-year trend charts. The deeper structure inside the data, for example which crops tend to be grown in the same season, which crops tend to appear together in the same district, and which crops tend to be linked with high or low yield, is rarely extracted. This structure is valuable because it describes how agriculture is actually practised on the ground, and it can be turned into simple guidance for planners, extension workers and farmers.

Association rule mining is a data mining technique that was first developed for market-basket analysis in retail, where it discovers which products customers buy together. The same idea transfers naturally to agriculture: a district in a given year and season can be treated as a "basket", and the crops grown there are the "products" in that basket. This project uses that idea to uncover crop patterns from more than 240,000 crop records.

1.2 Problem Definition

The problem addressed in this project is stated as follows: **given district-level crop production records for India from 1997 to 2015, discover the frequent and statistically meaningful relationships between crops, seasons, cultivated area, yield level and co-cultivated crops, and express them as interpretable if-then rules that can support crop planning and diversification decisions.**

The problem has two parts. The first part concerns the attributes of a single record: when a particular crop appears in a record, what can be said about its season, the size of the cultivated area and the level of yield? The second part concerns groups of crops: when certain crops are grown together in a district and season, which other crops are likely to be grown there too?

1.3 Objectives

The objectives of the project are:

1. To clean and prepare the national crop production data so that it is suitable for association rule mining.
2. To identify frequent crop, season, region and productivity patterns in the data.
3. To discover groups of crops that are frequently cultivated together in the same district, year and season (co-cultivation patterns).
4. To find associations between crops, seasons, cultivated area level and yield level.
5. To rank the discovered rules using support, confidence and lift, and to filter out weak or chance associations.

1.4 Scope of the Project

The work covers the following scope. The data used is the district-wise Crop Production in India file covering the years 1997 to 2015, with 124 crops, 33 states and union territories, 646 districts and six cropping seasons. Two complementary mining analyses are carried out: an attribute-based analysis in which every record is a transaction, and a basket-based analysis in which every district-year-season combination is a transaction. The project is limited to the attributes present in the dataset. It does not use rainfall, soil, temperature, irrigation or price data, and it does not attempt to forecast future production. The rules describe associations observed in the past data and should not be read as proof of cause and effect.

1.5 Significance of the Project

The results of this project can help several groups of users. Because the output is a set of short if-then rules with measured strength, it is easy to explain to people who are not data specialists.

- **Farmers and extension workers** can see which crops are commonly grown together and in which season, which supports decisions about intercropping and crop rotation.
- **Agricultural planners and policymakers** can identify crop clusters and regional cropping systems that already exist, and plan procurement, storage and input supply around them.
- **Researchers and students** obtain a documented, reproducible example of applying association rule mining to a real, large agricultural dataset.

2. Dataset

2.1 Source and Description

The project uses the publicly available Crop Production in India dataset, supplied as the file crop_production.csv. It is a district-level table in which each row records the area under one crop and the production of that crop in one district, in one year and in one season. The data covers the years 1997 to 2015. Crop statistics of this kind are normally compiled by state and district authorities and then collected into national tables, which is why the file has a regular structure with the same seven columns for every row.

The file contains 246,091 records and seven attributes. Four of the attributes (state, district, season and crop) are categorical, one (crop year) is a calendar year, and two (area and production) are numeric measurements. This combination makes the data well suited to association rule mining after the numeric attributes are converted into categories.

2.2 Attribute Description

Table 1: Attributes of the dataset

Attribute	Type	Description	Distinct values
State_Name	Categorical	State or union territory in which the record was reported	33
District_Name	Categorical	District in which the record was reported	646
Crop_Year	Numeric (year)	Calendar year of the record (1997 to 2015)	19
Season	Categorical	Cropping season: Kharif, Rabi, Whole Year, Summer, Winter or Autumn	6
Crop	Categorical	Name of the crop, for example Rice, Wheat, Maize, Sugarcane	124
Area	Numeric	Area under the crop, as recorded in the dataset	Continuous
Production	Numeric	Quantity produced, as recorded in the dataset	Continuous

2.3 Dataset Summary

Table 2: Summary of the dataset

Property	Value
Number of records	246,091
Number of attributes	7
States and union territories	33
Districts	646
Distinct crops	124
Seasons	6
Period covered	1997 to 2015 (19 years)
Missing values	3,730 in Production; none in the other six attributes
Records after cleaning	242,361

The two numeric attributes are strongly skewed, as shown in Table 3. The median area is only 582 while the largest area is 8,580,100, and the median production is 729 while the largest production is more than 1.25 billion. A small number of very large records therefore dominate the averages, which is the reason why the project converts the numeric attributes into rank-based categories instead of using raw values.

Table 3: Descriptive statistics of the numeric attributes

Statistic	Area	Production
Count	246,091	242,361
Mean	12,002.8	582,503.4
Standard deviation	50,523.4	17,065,810
Minimum	0.04	0
25th percentile	80	88
Median	582	729
75th percentile	4,392	7,023
Maximum	8,580,100	1,250,800,000

2.4 Characteristics and Limitations of the Dataset

Before mining, it is important to understand what the dataset can and cannot represent. The following points affect how the results should be read.

- **Uneven coverage.** Some states contribute far more records than others. Uttar Pradesh alone has 33,306 records, whereas Goa has only 208 and Chandigarh only 90. Patterns of large states therefore carry more weight in any national analysis.

- **Crop-specific units.** Production is recorded in the unit used for each crop, and these units are not the same for all crops. Raw production or yield values are therefore not strictly comparable across crops.
- **Grouped crop categories.** Some entries combine several crops, for example categories such as Other Rabi pulses, which hides differences between the individual crops inside them.
- **Partial years.** The year 2015 has only 562 records and 1997 has fewer records than later years, so the first and last years are less complete than the middle of the period.
- **Missing production values.** A small share of records (3,730, or 1.5 percent) has no production value.
- **No environmental variables.** Rainfall, soil type, temperature, irrigation and price information are not present, so the rules can describe what is grown together but not why.

3. Data Understanding and Preprocessing

3.1 Data Understanding

The first step was an exploratory look at the data to learn how records are distributed across seasons, crops, states and years. This guides the choice of mining thresholds later, because the share of records that belongs to a season or a crop sets an upper limit on the support that any rule involving it can reach.

Season distribution

Kharif is the most common season in the data, followed by Rabi and Whole Year. Summer, Winter and Autumn together account for about one tenth of all records. Table 4 and Figure 1 show the distribution.

Table 4: Number of records by season

Season	Records	Share of all records
Kharif	95,951	39.0%
Rabi	66,987	27.2%
Whole Year	57,305	23.3%
Summer	14,841	6.0%
Winter	6,058	2.5%
Autumn	4,949	2.0%

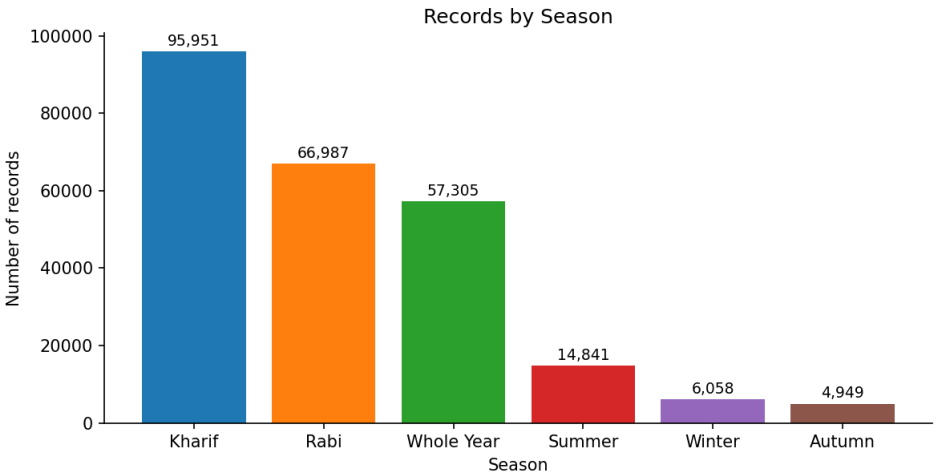


Figure 1: Number of records by season

Crop distribution

The dataset contains 124 different crops, but the records are concentrated in a few of them. Rice (15,104 records) and Maize (13,947) are the most frequently recorded, followed by Moong or Green Gram (10,318), Urad (9,850), Sesamum (9,046), Groundnut (8,834),

Sugarcane (7,921) and Wheat (7,899), as shown in Figure 2. Many other crops appear far less often, which means that rules involving rare crops can only appear if the minimum support is set low.

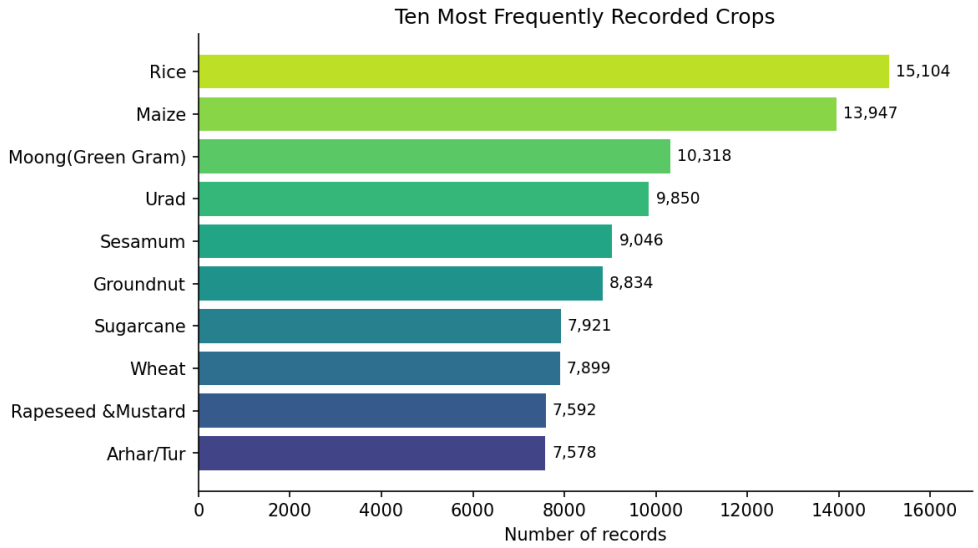


Figure 2: Ten most frequently recorded crops

State distribution

Figure 3 shows the ten states with the most records. Uttar Pradesh (33,306), Madhya Pradesh (22,943), Karnataka (21,122), Bihar (18,885), Assam (14,628) and Odisha (13,575) lead the list. Because the 33 states and union territories share the records among themselves, no single state exceeds 13.7 percent of the transactions in the attribute analysis.

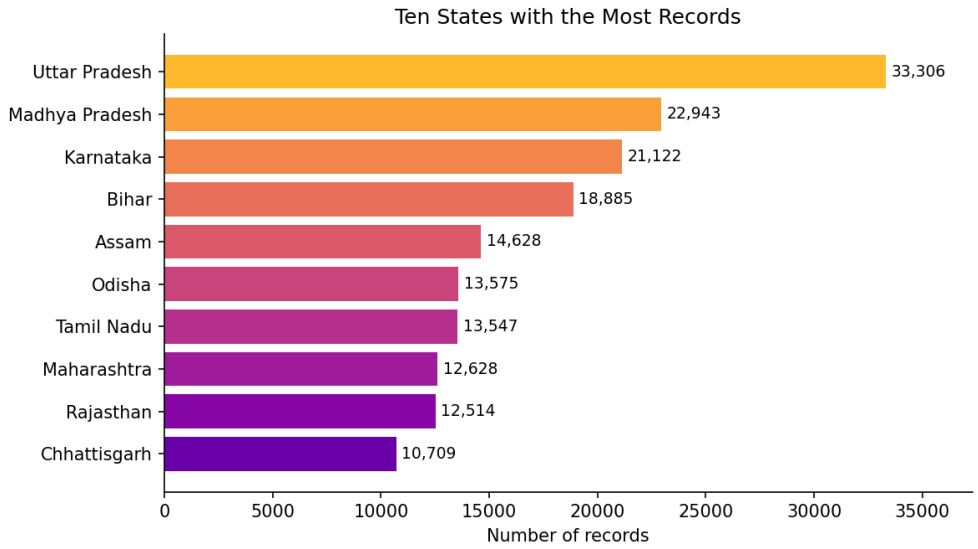


Figure 3: Ten states with the most records

Distribution over the years

Most years have roughly 11,000 to 17,000 records, with the highest count in 2003 (17,287). The first year, 1997, has 8,899 records, and 2015 is clearly incomplete with only 562 records (Figure 4). Since the analysis does not model time, these differences have only a small effect, but they are noted as a limitation.

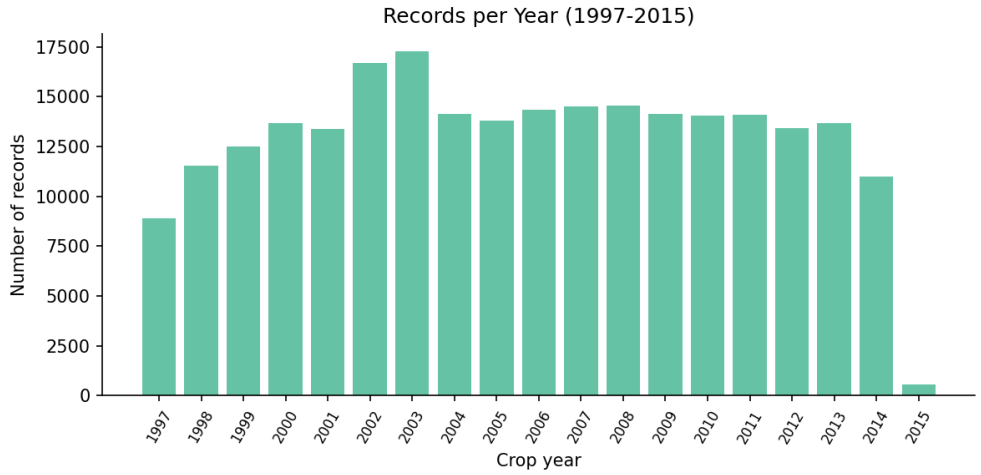


Figure 4: Number of records per year

3.2 Data Quality and Missing Values

A check of missing values showed that only the Production column is incomplete, with 3,730 missing entries out of 246,091 (about 1.5 percent). State, district, year, season, crop and area are fully populated. The check also showed that the Season column contains trailing blank spaces in its values, for example "Kharif" is stored followed by several spaces. If left unchanged, such values would be treated as different from the same word without spaces and would split a single season into several items.

3.3 Data Cleaning

Three cleaning operations were applied.

1. **Whitespace removal.** Leading and trailing spaces were removed from the State_Name, District_Name, Season and Crop columns so that every category has exactly one spelling.
2. **Removal of missing production.** The 3,730 records without a production value were removed, because a yield cannot be calculated without production. This left 242,361 records, which is 98.5 percent of the original data.
3. **Validity check on area.** Records with an area of zero would cause a division by zero when yield is calculated. The check found no such records, so none were removed.

3.4 Feature Creation: Yield

The raw columns Area and Production are both measures of size: a large district produces more simply because it cultivates more. To obtain a measure of productivity that does not depend on size, a new attribute called Yield was created as the ratio of production to area:

Yield = Production / Area

The yield attribute allows the analysis to ask questions such as which crops tend to give high or low output per unit of area, which cannot be answered from area or production alone.

3.5 Discretisation of Numeric Attributes

Association rule mining works with items, which are categorical, so the numeric attributes Area and Yield had to be converted into categories. Each of them was divided into three groups named Low, Medium and High.

The grouping was done on the rank of the values, which splits the records into three groups of equal size, each holding one third (33.3 percent, or about 80,787 records) of the cleaned data. Equal-sized groups were chosen for two reasons. First, because the values are extremely skewed (Table 3), equal-width intervals would put almost every record in the lowest group. Second, equal-sized groups give each level the same base support, so a rule linking a crop to the High level is not favoured or penalised simply because the High group is larger or smaller than the others.

Production was deliberately not converted into an item. Since production is largely determined by area, including both would generate many trivial rules of the type "high area implies high production" without adding agricultural insight. Yield, which is independent of size, was kept instead.

3.6 Summary of Preprocessing

Table 5: Summary of the preprocessing steps

Step	Action	Reason	Effect
1	Trimmed spaces in State_Name, District_Name, Season and Crop	Padded values such as Kharif followed by blanks would be counted as separate items	One spelling per category
2	Removed records with missing Production	Yield cannot be computed without production	246,091 to 242,361 records
3	Checked for zero Area	Avoid division by zero in the yield formula	No records removed
4	Created Yield = Production / Area	Size-independent measure of productivity	New attribute
5	Binned Area and Yield into Low, Medium, High (equal-sized groups on ranks)	Rules need categorical items; the numeric values are highly skewed	About 80,787 records per level
6	Excluded Production from the item set	Redundant with Area; would create trivial rules	Fewer, more useful rules

4. Predictive Modelling Approach

4.1 Association Rule Mining

Association rule mining searches a collection of transactions for itemsets that occur together more often than would be expected by chance, and then expresses these co-occurrences as rules of the form A implies B, written A → B. Here A is called the antecedent and B the consequent. A rule such as {Wheat} → {Rabi} states that records containing wheat also tend to contain the Rabi season. The technique does not need a target variable; it discovers any strong relationship present in the data, which is why it is called an unsupervised or descriptive method.

In the market-basket setting the transactions are shopping baskets and the items are products. In this project the same terms are mapped to agriculture. In the first analysis a transaction is a single crop record and the items are its attribute values. In the second analysis a transaction is a district in a given year and season, and the items are the crops grown there.

4.2 Rule-Quality Measures

Many rules can be generated from any dataset, so measures are needed to separate meaningful rules from accidental ones. Three main measures were used for filtering and ranking, and two further measures were recorded for additional insight.

Table 6: Rule-quality measures used in the project

Measure	Definition	Interpretation
Support	Support(A and B) = number of transactions containing both A and B / total transactions	How common the pattern is. A high value means the rule applies to many records.
Confidence	Confidence(A to B) = Support(A and B) / Support(A)	The proportion of transactions containing A that also contain B. It is the reliability of the rule.
Lift	Lift(A to B) = Confidence(A to B) / Support(B)	How many times more often A and B occur together than expected if they were independent. A value above 1 shows a positive association.
Leverage	Support(A and B) - Support(A) x Support(B)	The difference between observed and expected co-occurrence; larger positive values mean a stronger association.
Conviction	(1 - Support(B)) / (1 - Confidence(A to B))	How much more often the rule would be wrong if A and B were independent; higher values mean a stronger rule.

Support alone is not enough. A rule involving two very common crops has high support even if the crops are unrelated, and a rule with high confidence may simply point to a consequent that appears in almost every transaction. Lift corrects for this because it compares the observed

co-occurrence with the co-occurrence expected from the individual frequencies. For this reason, rules were required to meet a minimum confidence and a minimum lift, and the final ranking was done by lift.

4.3 Why Association Rule Mining for Crop Data

Several features of the problem make association rule mining a suitable technique. The data consists mainly of categorical attributes and of items that are either present or absent in a district, which fits the transaction model directly. The questions asked, such as which crops go together or which season a crop belongs to, are questions about co-occurrence and not about predicting a numeric value. The output is a list of short rules that anyone can read, which is an advantage over methods whose results are difficult to explain. Finally, the technique scales well to the size of the data, because efficient algorithms exist that avoid examining every possible combination of items.

4.4 Predictive Use of Association Rules

Association rule mining is descriptive, so there is no training set, no test set and no target column in this project. The discovered rules can still be used in a predictive way. Given a set of observed items, the rules whose antecedent matches those items indicate which other items are likely to be present, and the confidence of the rule gives the estimated probability. For example, if a record involves wheat, the rule {Wheat} → {Season = Rabi} with a confidence of 0.953 suggests that the season is Rabi with a probability of about 95 percent. In the same way, if a district grows arecanut, the rule {Arecanut} → {Coconut} with a confidence of 0.814 suggests that coconut is also likely to be grown there. This if-then reading is the predictive element of the approach.

4.5 Overall Methodology

The approach consists of seven stages, shown in Figure 5. The output of each stage is the input to the next.

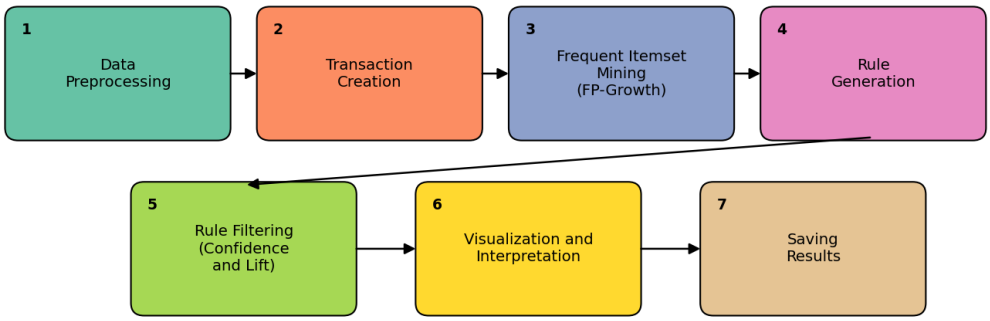


Figure 5: Methodology of the project

1. **Data preprocessing.** The raw file is cleaned, the yield attribute is created and the numeric attributes are converted into Low, Medium and High groups (Section 3).
2. **Transaction creation.** The cleaned table is converted into transactions in two different ways, one for each analysis (Sections 5.2 and 5.3).
3. **Frequent itemset mining.** The FP-Growth algorithm finds all itemsets whose support is at least the minimum support threshold.
4. **Rule generation.** For each frequent itemset, all possible splits into an antecedent and a consequent are formed and their confidence is calculated.
5. **Rule filtering.** Only rules that meet the minimum confidence and a minimum lift greater than one are kept, so that weak and chance associations are removed.
6. **Visualization and interpretation.** The rules are plotted and ranked, and the strongest rules are interpreted in agricultural terms.
7. **Saving results.** The frequent itemsets, the rules and the plots are written to files in a local results folder.

4.6 Choice of Algorithm

Two algorithms are widely used to find frequent itemsets: Apriori and FP-Growth. Apriori builds candidate itemsets level by level and scans the data repeatedly to count them, which becomes slow when the number of transactions and items is large. FP-Growth first compresses the transactions into a compact tree structure called an FP-tree and then extracts the frequent itemsets from the tree without generating candidates. Both algorithms return exactly the same frequent itemsets for the same threshold, but FP-Growth is considerably faster on large data. With more than 242,000 transactions and 169 items in the first analysis, FP-Growth was selected.

5. Model Development

5.1 Tools and Environment

The model was developed as a single Python program that reads the data file, runs both analyses and writes the results to disk. The following libraries were used.

- **pandas** for reading the file, cleaning the data, creating the yield attribute, grouping records into baskets and one-hot encoding.
- **mlxtend** for the FP-Growth algorithm (fpgrowth) and for rule generation (association_rules).
- **matplotlib** for the support-confidence scatter plots, which are saved as image files.

The program can be run on a local computer or on a cloud notebook such as Google Colab. The only settings that need to be changed are the path of the input file and the name of the output folder.

5.2 Transaction Design: Analysis A

In Analysis A every row of the cleaned data becomes one transaction. The items in a transaction are the values of five attributes: state, season, crop, the Low, Medium or High group of area, and the Low, Medium or High group of yield. Each item is written in the form attribute = value. For example, a record of wheat grown in Punjab in the Rabi season on a large area with a medium yield becomes the transaction {State = Punjab, Season = Rabi, Crop = Wheat, Area = High, Yield = Medium}.

There are 242,361 such transactions and 169 distinct items: 33 states, 6 seasons, 124 crops, 3 area groups and 3 yield groups. Every transaction contains exactly five items. Rules found in this analysis relate the attributes of a record to one another, such as a crop to a season or a crop to a yield level.

5.3 Transaction Design: Analysis B

In Analysis B the unit of observation is a district in a given year and season. All crop records that share the same state, district, year and season are collected into one basket, and the items of the basket are the crops grown there. In the market-basket analogy, the district-year-season is the shopper and the crops are the products bought.

This construction produced 34,955 baskets covering 124 crops. Baskets that contain only one crop cannot yield a rule, since a rule needs at least one item on each side, so these were

removed. This left 29,175 baskets with two or more crops, which were used for mining. The rules found in this analysis describe which crops are grown together.

5.4 One-Hot Encoding

Both sets of transactions were converted into a one-hot (True or False) table with one row per transaction and one column per item. A cell is True when the item is present in the transaction. For Analysis A the table was created from the five item columns, giving 242,361 rows and 169 columns. For Analysis B a cross-tabulation of baskets against crops gave a table of 34,955 rows and 124 columns, which was reduced to 29,175 rows after the single-crop baskets were removed. This is the input format required by the FP-Growth implementation.

Table 7: Comparison of the two transaction designs

	Analysis A: attribute rules	**Analysis B: co-cultivation rules**
One transaction is	One record (row) of the cleaned data	One basket: state, district, year and season
Items	State, season, crop, area group, yield group	Crops grown in the basket
Example of an item	Crop = Wheat; Season = Rabi; Yield = High	Wheat; Rapeseed and Mustard; Gram
Number of transactions used	242,361	29,175 (from 34,955 baskets)
Number of distinct items	169	124
Typical question answered	Which season, area level or yield level goes with a crop?	Which crops are grown together?

5.5 Frequent Itemset Mining

The FP-Growth algorithm was applied to each one-hot table. It works in two phases. In the first phase it scans the transactions to find the items whose individual support meets the threshold, and orders these items by frequency. In the second phase it builds an FP-tree in which transactions that share the same leading items share the same path, so the whole data set is stored compactly. The frequent itemsets are then obtained by following the paths of this tree, which avoids the creation of a very large number of candidate itemsets. A maximum itemset length of four was imposed, because rules with more than four items are difficult to interpret and rarely useful in practice.

5.6 Rule Generation and Filtering

From each frequent itemset with at least two items, every possible division into an antecedent and a consequent was considered. For an itemset of three items, for example, this gives six candidate rules. The confidence of each candidate was calculated from the supports that FP-Growth had already found, and candidates below the minimum confidence were discarded. The remaining rules were then filtered by lift, and only those with a lift of at least 1.2 were kept. A lift threshold above one removes rules in which the items are independent or negatively associated, and the value of 1.2 adds a margin so that only clear positive associations remain. The rules were finally sorted by lift and then by confidence.

5.7 Parameter Selection

The thresholds were chosen by considering the size and structure of each analysis. They are shown in Table 8.

Table 8: Mining parameters

Parameter	Analysis A	Analysis B	Reason
Minimum support	0.02	0.03	A itemset must appear in at least 2 percent (A) or 3 percent (B) of transactions. B has far fewer transactions, so a slightly higher value keeps the rule count manageable.
Minimum confidence	0.60	0.60	At least 60 percent of the transactions containing the antecedent must also contain the consequent.
Minimum lift	1.2	1.2	Keeps only clear positive associations and removes chance co-occurrence.
Maximum itemset length	4	4	Keeps rules short enough to be read and interpreted.

The support of 0.02 for Analysis A is low enough to let moderately frequent crops such as Rice, Maize, Wheat and Sugarcane take part in the rules, but high enough to avoid the very large number of rare itemsets that a smaller value would create. Because the 124 crops and 33 states divide the records among many items, even the most common single crop (Rice) has a support of only 0.062 in this analysis. Setting the threshold much higher would therefore remove almost all crop-based rules.

5.8 Output Files

For each analysis the program saves three files to the results folder, so that the complete output can be examined, reported and reused without running the model again.

Table 9: Files saved by the program

File	Content
A_attributes_frequent_itemsets.csv	All frequent itemsets of Analysis A with their support and length
A_attributes_rules.csv	The 33 rules of Analysis A with support, confidence, lift, leverage, conviction and related measures, sorted by lift
A_attributes_rules_plot.png	Scatter plot of support against confidence for Analysis A, coloured by lift
B_cocultivation_frequent_itemsets.csv	All frequent itemsets of Analysis B with their support and length
B_cocultivation_rules.csv	The 12,569 rules of Analysis B with the same measures, sorted by lift
B_cocultivation_rules_plot.png	Scatter plot of support against confidence for Analysis B, coloured by lift

5.9 Implementation Summary

The complete procedure carried out by the program can be summarised in the following steps.

1. Read crop_production.csv, remove padding spaces, drop records with missing production and compute the yield.
2. Divide area and yield into Low, Medium and High groups of equal size.
3. Build the transactions and one-hot tables for Analysis A (record level) and Analysis B (district-year-season baskets).
4. Run FP-Growth on each table with the minimum support and the maximum itemset length of Table 8.
5. Generate rules, filter them by confidence and lift, and sort them by lift.
6. Convert the itemsets and rules to readable text, save them as CSV files and draw the support-confidence plots.
7. Print the top rules of each analysis for a quick review.

6. Evaluation of the Model

6.1 Evaluation Strategy

Association rule mining has no ground-truth labels, so the accuracy and error measures used for classification models do not apply. The quality of the model was instead judged in four ways: by the number and size of the frequent itemsets, by the statistical strength of the rules (support, confidence, lift, leverage and conviction), by the visual distribution of the rules in the support-confidence plots, and by checking whether the strongest rules agree with established agricultural knowledge. The reliability of the thresholds was also considered.

6.2 Frequent Itemsets

Analysis A found 180 frequent itemsets, made up of 43 single items, 105 pairs and 32 triples. No itemset of four items reached the minimum support of 0.02. Analysis B found 3,658 frequent itemsets, made up of 48 single crops, 371 pairs, 1,089 triples and 2,150 itemsets of four crops. Table 10 and Figure 6 show these counts.

Table 10: Frequent itemsets by size

Itemset size	Analysis A	Analysis B
1 item	43	48
2 items	105	371
3 items	32	1,089
4 items	0	2,150
Total	180	3,658

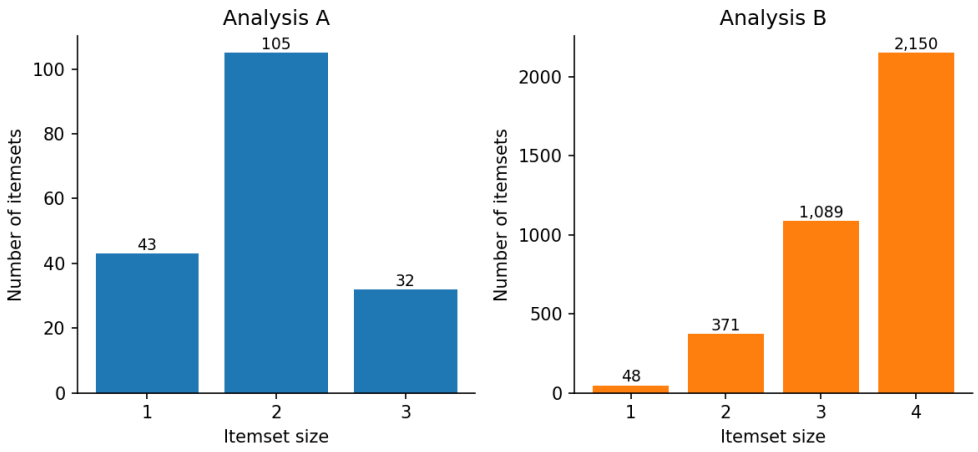


Figure 6: Number of frequent itemsets by size for the two analyses

The two analyses behave very differently. In Analysis A the number of itemsets falls quickly as the size increases, because each transaction holds five items from very different groups

(state, season, crop, area and yield) and few combinations of three or more are common enough. In Analysis B the number of itemsets rises with size up to four items. The crops grown in a district form tightly linked groups, so once a few crops of a typical cropping system are present, many combinations of them are also frequent. The average support also falls with the size of the itemset, as expected: in Analysis A it is 0.103 for single items, 0.049 for pairs and 0.033 for triples, and in Analysis B it is 0.161, 0.078, 0.060 and 0.051 for itemsets of one to four crops.

6.3 Rule Statistics

After filtering by confidence and lift, 33 rules remained in Analysis A and 12,569 in Analysis B. Table 11 summarises their quality measures.

Table 11: Statistics of the generated rules

Statistic	Analysis A	Analysis B
Number of rules	33	12,569
Support: minimum to maximum	0.021 to 0.050	0.030 to 0.311
Support: median	0.026	0.050
Confidence: minimum to maximum	0.601 to 0.991	0.600 to 1.000
Confidence: median	0.787	0.801
Confidence: mean	0.794	0.795
Lift: minimum to maximum	1.55 to 7.48	1.36 to 18.20
Lift: median	2.48	3.35
Lift: mean	2.79	3.62
Leverage: mean	0.0168	0.0402
Conviction: median	3.46	3.68

In both analyses the average confidence is close to 0.80, which means that, on average, a rule is correct in about four out of five of the transactions in which its antecedent appears. Every rule has a lift above 1.35, and the median lift is 2.48 for Analysis A and 3.35 for Analysis B, so the typical rule describes items that occur together two to three times more often than chance would produce. The mean leverage is positive in both analyses, and the median conviction of about 3.5 to 3.7 shows that the rules would be wrong far more often if the items were independent. In Analysis A eight rules have a confidence of at least 0.90, and in Analysis B 2,652 rules have a confidence of at least 0.90, 977 have a confidence of at least 0.95 and 27 have a confidence of 1.00.

6.4 Support-Confidence Plots

Figures 7 and 8 plot every rule by its support (horizontal axis) and confidence (vertical axis), with the colour showing the lift. They were saved by the program as part of the output.

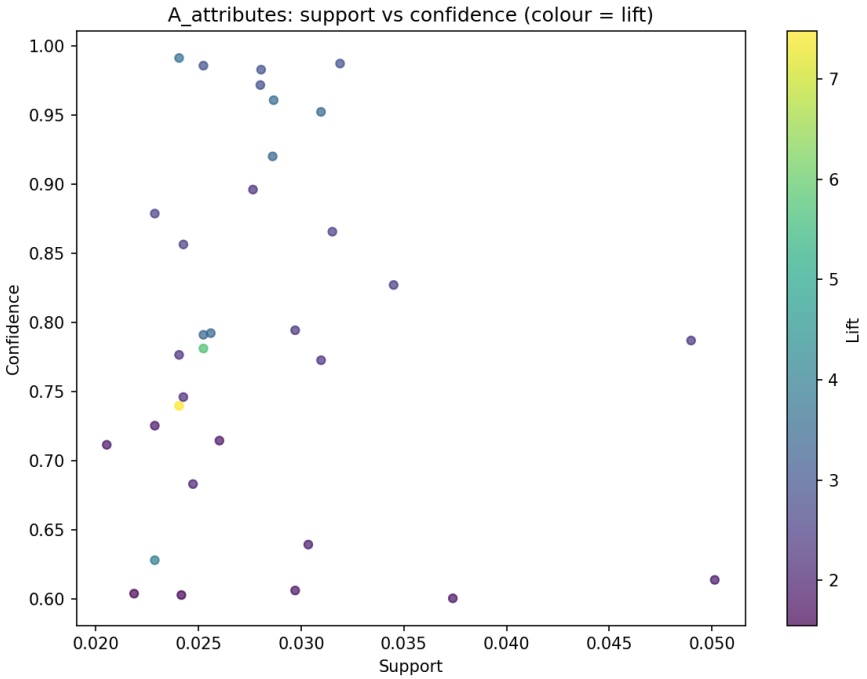


Figure 7: Support versus confidence for Analysis A (colour shows lift)

In Analysis A the 33 rules are spread over supports from 0.02 to 0.05 and over confidences from 0.60 to 0.99. Most points are dark purple to blue, which corresponds to lifts of about 2 to 4. A single rule stands out in yellow with the highest lift of about 7.5, and one green point has a lift near 5.7. The rules with the highest confidence, above 0.95, are also those with a moderately high lift, which shows that they are not simply driven by a very common consequent. The rule set is small, so each rule can be examined individually.

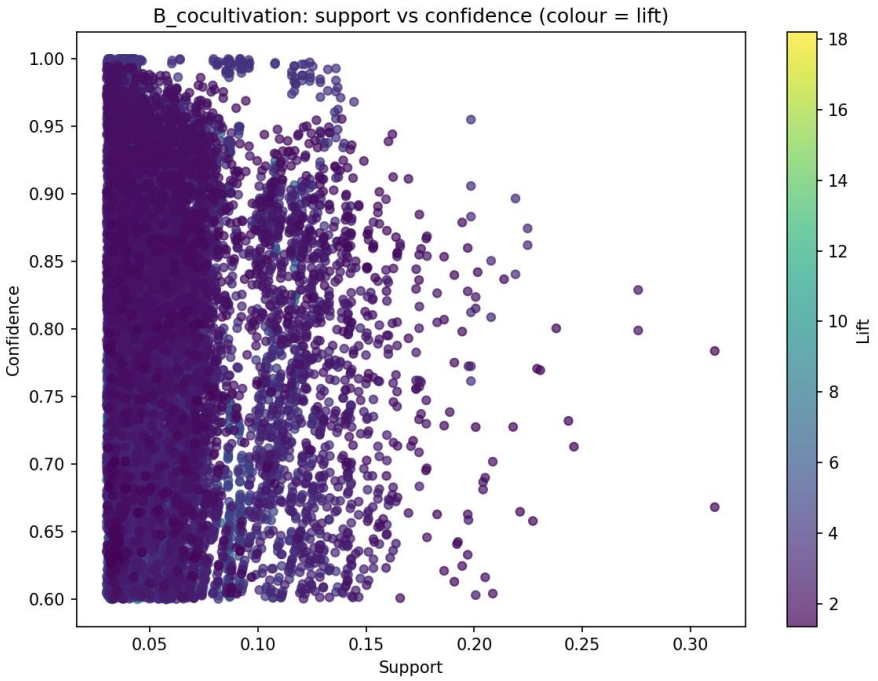


Figure 8: Support versus confidence for Analysis B (colour shows lift)

In Analysis B the picture is quite different. Most of the 12,569 rules are packed into the narrow band of support between 0.03 and 0.10 and spread over the whole range of confidence from 0.60 to 1.00. In total, 11,137 rules have a support below 0.10, and only 30 rules have a support above 0.20. The rules with the highest support, reaching 0.31, have a dark colour, which means a low lift, because they involve crops such as Maize and Rice that are grown almost everywhere. The brighter, high-lift rules are located at low support, where they describe specific crop combinations. This shows the expected trade-off: the more specific and surprising a rule is, the fewer districts it applies to.

6.5 Distribution of Lift and Confidence

The distribution of the rules over lift intervals is shown in Table 12. In Analysis B, nearly half of the rules (6,079) have a lift between 3 and 5, and 1,694 rules have a lift of 5 or more, including 59 rules with a lift of 10 or more. In Analysis A, 24 of the 33 rules have a lift between 1.2 and 3.

Table 12: Number of rules by lift interval

Lift interval	Analysis A	Analysis B
1.2 to 2	9	599
2 to 3	15	4,197
3 to 5	7	6,079
5 to 10	2	1,635
10 and above	0	59

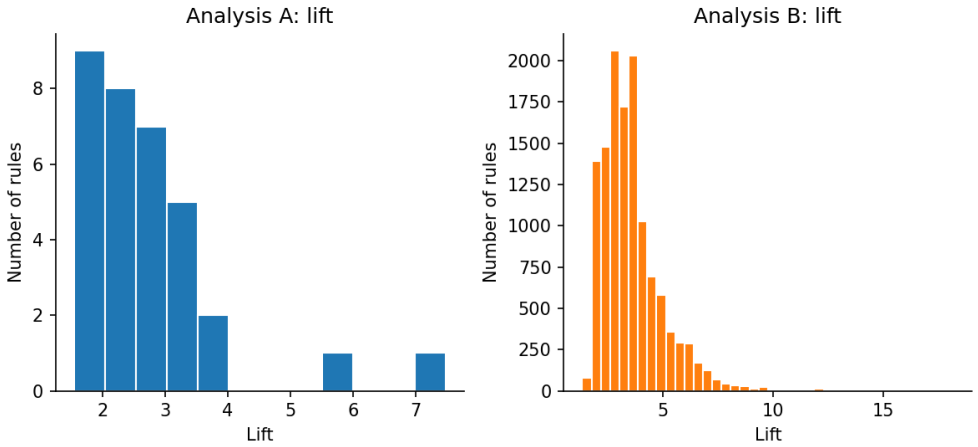


Figure 9: Distribution of lift for the two analyses

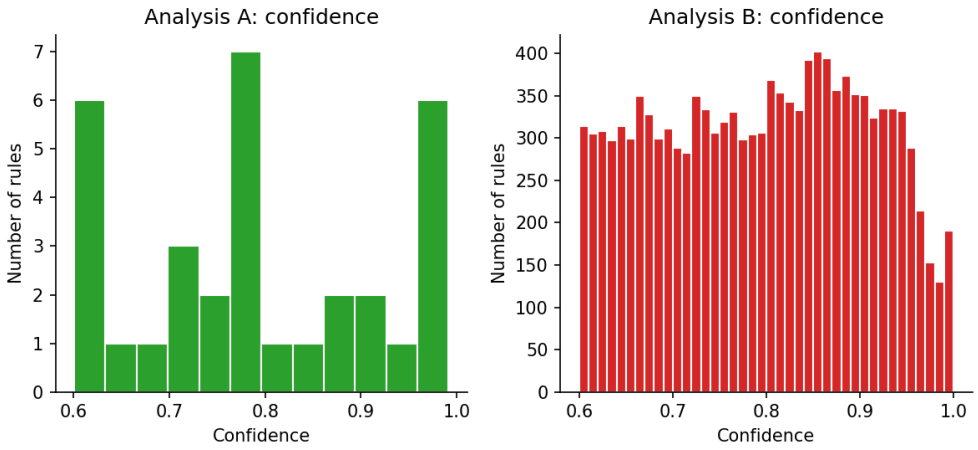


Figure 10: Distribution of confidence for the two analyses

The lift histogram of Analysis B is roughly bell-shaped with a long right tail: about 82 percent of the rules have a lift between 2 and 5, and the tallest bar lies near 2.6 to 3.0. The confidence histogram shows that about 67 percent of the rules have a confidence between 0.70 and 0.95, with the tallest bar near 0.85, and a small number reach 1.00. Rules with a confidence of 1.00 are those in which the consequent appeared every time the antecedent appeared. An example

is the rule {Gram, Khesari, Rapeseed and Mustard} → {Wheat}, which has a confidence of 1.00 and a lift of 3.84.

6.6 Validation Against Agricultural Knowledge

Since no labelled answer exists, the results were checked against well-known facts of Indian agriculture. The strongest rules agree with these facts, which supports the validity of the mined patterns.

- Wheat, gram and rapeseed or mustard are linked to the Rabi season with confidences between 0.92 and 0.96, while arhar or tur is linked to the Kharif season, as expected from their usual sowing calendars.
- Arecanut, coconut, banana, tapioca and turmeric, which are plantation and horticultural crops, appear together in the rules with some of the highest lifts.
- Jute and mesta, two fibre crops, are found to be strongly linked with each other.
- Wheat and rapeseed or mustard, which are Rabi cereals and oilseeds that are commonly grown together, are strongly associated in the co-cultivation analysis.

No rule was found that contradicts common agricultural knowledge. Some surprising rules, such as the links between several Rabi pulses and wheat, are consistent with the well-known practice of growing pulses and cereals within the same Rabi cropping system.

6.7 Reliability and Sensitivity

The results depend on the thresholds, and the following points should be kept in mind when they are interpreted.

- **Support threshold.** A lower minimum support would add rules about rarer crops, and a higher one would keep only the most widespread patterns. The values chosen allow 17 crops to appear as single frequent items in Analysis A and 48 crops in Analysis B.
- **Redundancy in Analysis B.** One frequent itemset of four crops produces many rules, because it can be split in several ways into antecedent and consequent. For this reason the number of rules (12,569) is much larger than the number of independent crop groups, and many rules describe the same cluster of crops.
- **No state-level rules in Analysis A.** None of the 33 rules contains a state, because the records are spread over 33 states and the state-level patterns fall below the minimum support. State-specific rules need a lower threshold or a separate analysis for each state.

- **Association is not causation.** A high lift shows that items occur together, not that one causes the other.
- **Yield groups.** The Low, Medium and High yield groups were computed for all crops together, and since production units differ between crops, yield-based rules partly reflect these units.

7. Results and Interpretation

7.1 Most Frequent Items and Itemsets

Before looking at the rules, it is useful to see which items and itemsets are the most frequent, because the rules are built from them.

Analysis A

Among the single items of Analysis A, the three most frequent are Season = Kharif (38.9 percent of all records), Season = Rabi (27.3 percent) and Season = Whole Year (23.2 percent). The three area groups and the three yield groups each have exactly 33.3 percent, since they were built from equal-sized groups. The most frequent crops are Rice (6.2 percent) and Maize (5.7 percent), followed by Moong (4.2 percent) and Urad (4.0 percent). Fourteen states reach the minimum support, led by Uttar Pradesh (13.7 percent), Madhya Pradesh (9.3 percent) and Karnataka (8.7 percent). The most frequent pairs of items are listed in Table 13.

Table 13: Most frequent itemsets of two items in Analysis A

Itemset	Support (% of records)
Season = Kharif, Yield = Low	15.9%
Area = High, Season = Kharif	15.0%
Season = Kharif, Yield = Medium	14.6%
Season = Whole Year, Yield = High	13.6%
Area = Medium, Season = Kharif	12.5%
Area = Medium, Yield = Low	12.1%
Area = High, Yield = High	12.1%
Area = High, Yield = Medium	11.6%
Area = Low, Yield = Low	11.6%
Area = Low, Season = Kharif	11.4%

Analysis B

In Analysis B, Maize is the most common crop and is present in 46.6 percent of the baskets. Rice follows with 39.7 percent, then Moong or Green Gram with 34.5 percent, Urad with 33.3 percent, Sesamum with 30.0 percent, Groundnut with 29.7 percent, Wheat with 26.1 percent, Rapeseed and Mustard with 25.7 percent and Arhar or Tur with 25.5 percent (Figure 11).

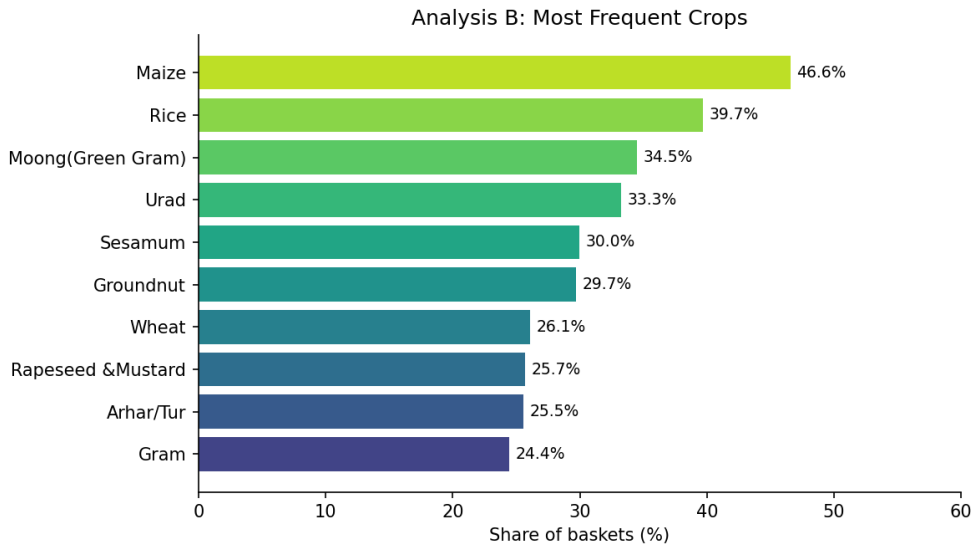


Figure 11: Most frequent crops in the baskets of Analysis B

The most common pair of crops is Maize and Rice, found together in 31.1 percent of the baskets, and the second is Moong and Urad with 27.6 percent. The most common sets of two and three crops are given in Tables 14 and 15.

Table 14: Most frequent pairs of crops in Analysis B

Crops	Support (% of baskets)
Maize, Rice	31.1%
Moong(Green Gram), Urad	27.6%
Maize, Moong(Green Gram)	24.6%
Maize, Urad	24.4%
Groundnut, Maize	23.8%
Maize, Sesamum	23.1%
Groundnut, Rice	22.9%
Moong(Green Gram), Rice	22.7%
Rapeseed &Mustard, Wheat	22.5%
Rice, Urad	22.1%

Table 15: Most frequent sets of three crops in Analysis B

Crops	Support (% of baskets)
Maize, Moong(Green Gram), Urad	20.1%
Gram, Rapeseed &Mustard, Wheat	19.8%
Groundnut, Maize, Rice	19.7%
Maize, Rice, Urad	19.4%
Maize, Moong(Green Gram), Rice	19.1%
Maize, Rice, Sesamum	18.6%
Moong(Green Gram), Rice, Urad	18.3%
Groundnut, Moong(Green Gram), Urad	17.8%

The most frequent sets of crops combine the staples Maize, Rice, Moong, Urad, Groundnut and Sesamum, which together form the typical Kharif cropping mix, and the set of Gram, Rapeseed and Mustard and Wheat, which forms the typical Rabi mix. Four crops, Maize, Moong, Rice and Urad, are found together in 16.0 percent of the baskets.

7.2 Analysis A: Crop, Season, Area and Yield Rules

Analysis A produced 33 rules. In 31 of them the antecedent contains a crop, which shows that the crop is the main driver of the patterns. Fifteen rules have a season in the consequent, fifteen have a yield group and six have an area group. The complete list is given in Table 16.

Table 16: All rules of Analysis A, sorted by lift

No.	Antecedent	Consequent	Support	Confidence	Lift
1	Crop = Wheat	Area = High, Season = Rabi	0.024	0.740	7.48
2	Crop = Sugarcane	Season = Whole Year, Yield = High	0.025	0.781	5.74
3	Crop = Sesamum	Season = Kharif, Yield = Low	0.023	0.628	3.96
4	Area = High, Crop = Wheat	Season = Rabi	0.024	0.991	3.63
5	Crop = Gram	Season = Rabi	0.029	0.961	3.52
6	Crop = Wheat	Season = Rabi	0.031	0.953	3.49
7	Crop = Sugarcane	Season = Whole Year	0.026	0.793	3.42
8	Crop = Sugarcane, Yield = High	Season = Whole Year	0.025	0.791	3.42
9	Crop = Rapeseed &Mustard	Season = Rabi	0.029	0.920	3.37
10	Crop = Sugarcane	Yield = High	0.032	0.987	2.96
11	Crop = Sugarcane, Season = Whole Year	Yield = High	0.025	0.986	2.96

No.	Antecedent	Consequent	Support	Confidence	Lift
12	Crop = Potato	Yield = High	0.028	0.983	2.95
13	Crop = Onion	Yield = High	0.028	0.972	2.92
14	Crop = Sesamum, Season = Kharif	Yield = Low	0.023	0.879	2.64
15	Crop = Sesamum	Yield = Low	0.032	0.866	2.60
16	Crop = Rice, Season = Kharif	Area = High	0.024	0.857	2.57
17	Crop = Moong(Green Gram)	Yield = Low	0.034	0.827	2.48
18	Crop = Rice, Yield = High	Area = High	0.030	0.795	2.38
19	Crop = Rice	Area = High	0.049	0.787	2.36
20	Crop = Wheat, Season = Rabi	Area = High	0.024	0.777	2.33
21	Crop = Urad	Yield = Low	0.031	0.773	2.32
22	Crop = Arhar/Tur	Season = Kharif	0.028	0.896	2.30
23	Crop = Wheat	Area = High	0.024	0.746	2.24
24	Crop = Groundnut	Yield = Medium	0.025	0.683	2.05
25	Area = High, Season = Whole Year	Yield = High	0.030	0.640	1.92
26	Crop = Sesamum, Yield = Low	Season = Kharif	0.023	0.726	1.87
27	Area = Medium, Season = Whole Year	Yield = High	0.050	0.614	1.84
28	Crop = Sesamum	Season = Kharif	0.026	0.715	1.84
29	Crop = Jowar	Season = Kharif	0.021	0.712	1.83
30	Area = High, Crop = Rice	Yield = High	0.030	0.606	1.82
31	Crop = Rice	Yield = High	0.037	0.601	1.80
32	Crop = Groundnut	Season = Kharif	0.022	0.604	1.55
33	Crop = Urad	Season = Kharif	0.024	0.603	1.55

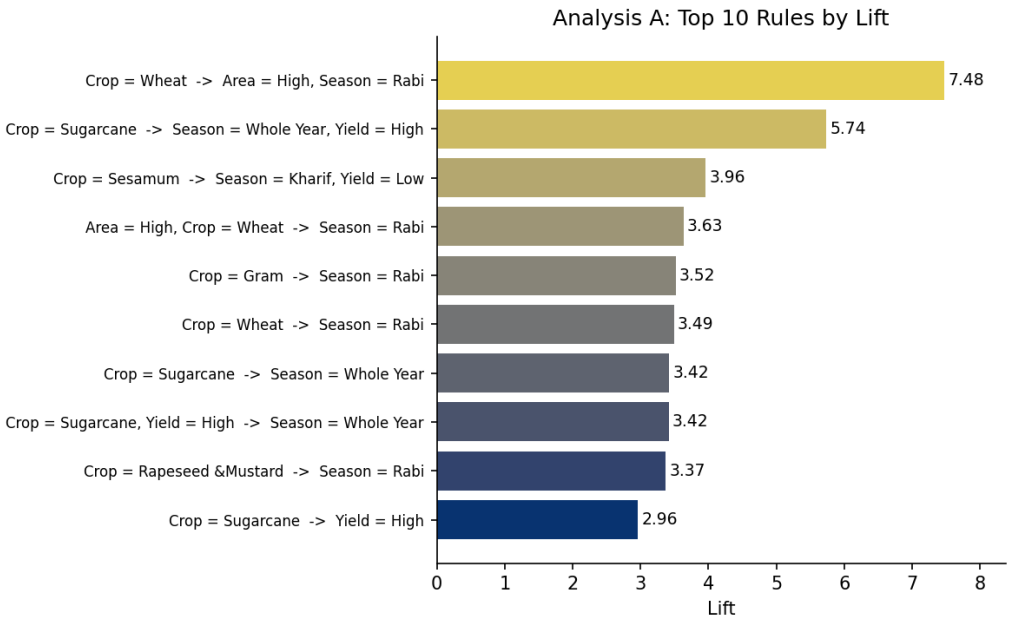


Figure 12: Top ten rules of Analysis A ranked by lift

Seasonal patterns

The clearest result is that crops are tied to seasons. Wheat implies the Rabi season with a confidence of 0.953 and a lift of 3.49. For wheat grown on a large area the confidence rises to 0.991, which means that almost every large-area wheat record belongs to the Rabi season. Gram (0.961), Rapeseed and Mustard (0.920) show the same strong link with Rabi. At the other end, Arhar or Tur is linked to Kharif with a confidence of 0.896, and Sesamum, Jowar, Groundnut and Urad are also linked to Kharif with confidences between 0.60 and 0.72. Sugarcane is linked to the Whole Year season (confidence 0.793, lift 3.42), which fits its long growth period.

Cultivated area patterns

The rule {Wheat} → {Area = High, Season = Rabi} has the highest lift of Analysis A at 7.48 and a confidence of 0.740, meaning that nearly three quarters of wheat records fall in the High area group and in the Rabi season at the same time. Rice is linked to the High area group with a confidence of 0.787, which rises to 0.857 when Kharif rice is considered. These rules reflect the position of wheat and rice as staple crops that are grown at large scale.

Yield patterns

Sugarcane (0.987), Potato (0.983) and Onion (0.972) are almost always found in the High yield group, while Sesamum (0.866), Moong (0.827) and Urad (0.773) are found mainly in the Low yield group. Rice is moderately linked to the High yield group (0.601). Groundnut is linked to the Medium yield group (0.683). It must be remembered that production units differ between

crops and that the yield groups were computed over all crops together. Crops that are measured in heavy units, such as sugarcane, potato and onion, will naturally appear in the High group, and the rules therefore indicate how recorded yield values compare across crops and not a true ranking of productivity.

7.3 Analysis B: Co-Cultivation Rules

Analysis B produced 12,569 rules. Their structure is shown in Table 21. The most common type (6,127 rules) has three crops in the antecedent and one in the consequent. There are 3,956 rules with two crops on each side and 1,934 rules with two crops in the antecedent and one in the consequent. Only 205 rules link a single crop to a single crop.

Widely grown combinations

The rules with the highest support involve the staple crops grown in many districts. They are listed in Table 17.

Table 17: Co-cultivation rules with the highest support

Antecedent	Consequent	Support	Confidence	Lift
Rice	Maize	0.311	0.784	1.68
Maize	Rice	0.311	0.668	1.68
Urad	Moong (Green Gram)	0.276	0.829	2.40
Moong (Green Gram)	Urad	0.276	0.799	2.40
Moong (Green Gram)	Maize	0.246	0.713	1.53
Urad	Maize	0.244	0.732	1.57
Groundnut	Maize	0.238	0.801	1.72
Sesamum	Maize	0.231	0.770	1.65
Groundnut	Rice	0.229	0.771	1.94
Moong (Green Gram)	Rice	0.227	0.658	1.66
Rapeseed and Mustard	Wheat	0.225	0.875	3.36
Wheat	Rapeseed and Mustard	0.225	0.862	3.36

In 78.4 percent of the baskets that contain Rice, Maize is also present, and in 82.9 percent of the baskets with Urad, Moong is present as well. The lifts of these rules are moderate, between 1.5 and 2.4, because the crops involved are so widespread that they co-occur often by chance too. The pair of Rapeseed and Mustard with Wheat is different: a lift of 3.36 together with a support of 0.225 shows a strong and also common Rabi cropping pattern.

Strongly linked combinations

The rules with the highest lift describe specific groups of crops that are far more likely to be found together than separately. They are given in Tables 18 and 19 and in Figure 13.

Table 18: Strong co-cultivation rules between single crops

Antecedent	Consequent	Support	Confidence	Lift
Arecanut	Coconut	0.043	0.814	12.40
Coconut	Arecanut	0.043	0.653	12.40
Jute	Mesta	0.034	0.710	11.71
Tapioca	Coconut	0.035	0.644	9.82
Cashewnut	Banana	0.031	0.853	7.96
Tapioca	Banana	0.043	0.788	7.35
Coconut	Banana	0.050	0.757	7.06
Khesari	Masoor	0.046	0.985	6.92
Garlic	Coriander	0.083	0.750	6.63
Coriander	Garlic	0.083	0.731	6.63

Table 19: Strong co-cultivation rules between groups of crops

Antecedent	Consequent	Support	Conf.	Lift
Masoor, Moong	Peas and beans, Urad	0.031	0.749	18.20
Peas and beans, Urad	Masoor, Moong	0.031	0.750	18.20
Masoor, Urad	Moong, Peas and beans	0.031	0.778	16.67
Tapioca, Turmeric	Banana, Coconut	0.031	0.754	15.18
Arecanut, Sugarcane	Coconut, Turmeric	0.033	0.823	14.61
Coconut, Sweet potato	Tapioca	0.030	0.780	14.38
Masoor, Moong	Urad, Wheat	0.031	0.758	13.93
Arecanut, Sugarcane	Coconut	0.035	0.892	13.59

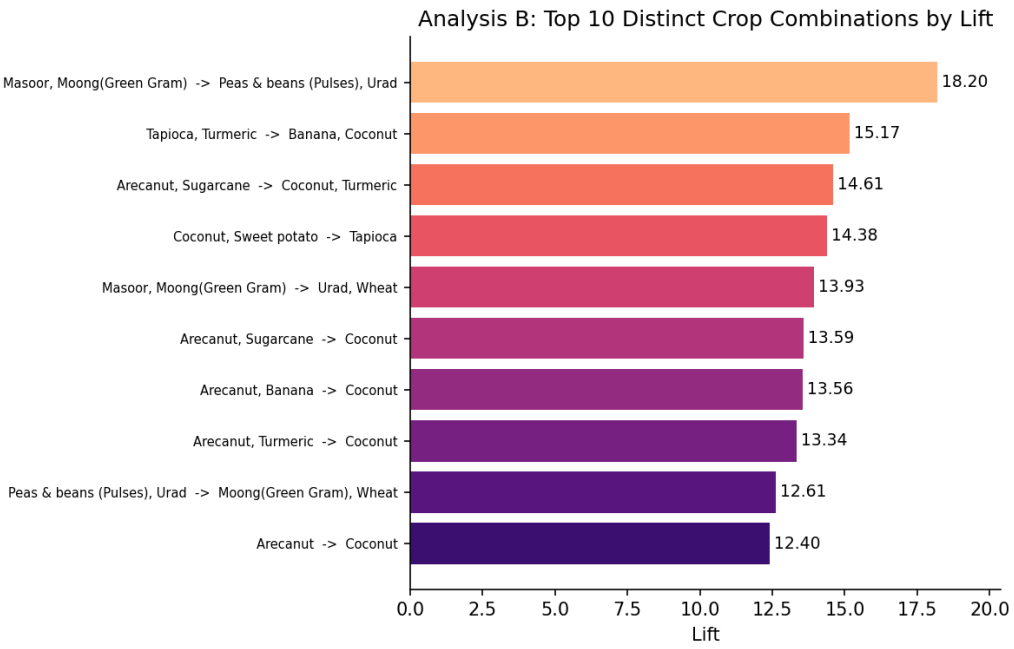


Figure 13: Top ten distinct crop combinations of Analysis B ranked by lift

Pulse clusters

Several pulses form tight groups. Khesari implies Masoor with a confidence of 0.985, and the sets {Khesari, Linseed, Peas and beans} → {Masoor} reach a confidence of 0.995. The rules with the highest lift of all, 18.20, link the group {Masoor, Moong} with {Peas and beans, Urad}. The same pulses are connected to Wheat, as in {Masoor, Moong} → {Urad, Wheat}. Together these rules show that pulses are commonly grown as a group, and that they are woven into the cereal-based Rabi system. The confidence of 1.00 for {Gram, Khesari, Peas and beans} → {Wheat} means that in every basket where those three pulses were present, wheat was grown as well.

Plantation and horticultural clusters

The crops Arecanut, Coconut, Banana, Tapioca, Turmeric, Cashewnut and Sweet potato form a second cluster with very high lifts, between 7 and 15. For instance, Arecanut implies Coconut with a confidence of 0.814 and a lift of 12.40, and the pair {Arecanut, Sugarcane} implies Coconut with a confidence of 0.892. These are typical plantation and horticultural cropping systems, in which perennial and semi-perennial crops share the same land or district. Because the state was not part of the baskets, the regions in which these clusters occur were not tested in this analysis.

Other crop pairs

Jute implies Mesta (lift 11.71), which fits the fact that both are fibre crops grown under similar conditions. Garlic and Coriander have a lift of 6.63 and a support of 0.083, which makes this pair both strong and relatively common, and suggests a spice and vegetable mix.

7.4 Comparison of the Two Analyses

The two analyses answer different questions and complement each other. Analysis A describes a single record and relates a crop to its season, area and yield. Analysis B describes a district and shows which crops are grown together. Table 20 compares them, and Table 21 shows the structure of the rules of Analysis B.

Table 20: Comparison of Analysis A and Analysis B

Aspect	Analysis A	Analysis B
Question answered	Which season, area group or yield group goes with a crop?	Which crops are grown together?
Transactions	242,361 records	29,175 baskets
Frequent itemsets	180	3,658
Rules	33	12,569
Highest lift	7.48	18.20
Median confidence	0.787	0.801
Main finding	Crops are tied to seasons; staples occupy large areas	Distinct crop clusters: staples, pulses, Rabi mix, plantation crops
Type of rule set	Small and easy to read	Large; needs filtering by lift or support

Table 21: Number of rules in Analysis B by size of antecedent and consequent

Crops in antecedent	1 crop in consequent	2 crops in consequent	3 crops in consequent
1	205	233	114
2	1,934	3,956	0
3	6,127	0	0

7.5 Practical Applications

The rules can be used in several practical ways.

- **Crop calendar support.** The seasonal rules provide a data-based calendar that shows when each crop is normally grown.

- **Intercropping and diversification.** Rules such as Rapeseed and Mustard with Wheat, or the pulse clusters, show which crops already work together in the same district, and these can be recommended to farmers who wish to diversify.
- **Input planning.** Once a crop is known to be grown in an area, the rules show which other crops are likely to be present, which helps in planning the supply of seeds, fertilisers and storage.
- **Regional planning.** Planners can identify existing cropping systems, such as the plantation cluster, and plan support for them.
- **Recommendation tool.** The rules can be put into a simple tool that suggests likely companion crops for a selected crop and season.

8. Conclusion

8.1 Summary of the Work

This project applied association rule mining with the FP-Growth algorithm to the district-level Crop Production in India dataset for 1997 to 2015. The raw file of 246,091 records was cleaned to 242,361 records, a yield attribute was created, and the numeric attributes were converted into Low, Medium and High groups. Two analyses were carried out. The attribute analysis treated every record as a transaction and produced 180 frequent itemsets and 33 rules that link crops to seasons, cultivated area and yield. The co-cultivation analysis treated every district-year-season as a basket of crops and produced 3,658 frequent itemsets and 12,569 rules that describe which crops are grown together, with lift values up to 18.2. The frequent itemsets, rules and plots were saved to files for further use.

8.2 Key Findings

- Crops follow strong seasonal patterns. Wheat, gram and rapeseed or mustard are almost entirely Rabi crops (confidence 0.92 to 0.96), arhar or tur is mainly a Kharif crop, and sugarcane is mostly a Whole Year crop.
- Staple crops such as wheat and rice are associated with large cultivated areas, and 85.7 percent of Kharif rice records are in the High area group.
- Maize and Rice are the most widely grown pair of crops, found together in 31.1 percent of the baskets, and Moong and Urad form the most common pair of pulses.
- Distinct crop clusters exist: a Kharif staple and pulse mix, a Rabi mix of wheat, gram and mustard, tightly linked pulse groups and a plantation and horticultural group of arecanut, coconut, banana, tapioca and turmeric.
- The strongest rules agree with established agricultural knowledge, which supports the usefulness of the approach for crop planning and diversification.

8.3 Limitations

- Association rules show co-occurrence and not cause and effect.
- Production is recorded in crop-specific units, and the yield groups were computed over all crops together, so yield-based rules partly reflect differences in units.
- The minimum support threshold removes rare crops from the results.

- Analysis B produces many overlapping rules, because each itemset can be divided into several antecedents and consequents.
- The year 2015 is only partially covered, and the time dimension was not modelled.
- No state-level rules were found in Analysis A, and no weather, soil or price information was used.

8.4 Future Scope

1. Mine rules separately for each state or agro-climatic region to obtain regional recommendations.
2. Normalise yield within each crop before grouping, so that the effect of different units is removed.
3. Combine the crop data with rainfall, temperature and soil datasets to link growing conditions to crop choice.
4. Prune redundant rules and use leverage and conviction together with lift to select the most useful rules.
5. Add a time dimension to study how crop combinations change over the years.
6. Build a simple interactive tool that suggests likely companion crops for a selected crop and season.